

Small Groups

The lexicon, and the three places the page and the paper do not line up

The index question for this project is two questions, and only the first has an answer yet.

“how do people reach consensus on deliberation tasks in small groups? how can we make it more efficient?”

website, Problem-Solving, the three settings

This is the one project in the Mental World Modeling cluster whose unit is not a mental operation. The unit is an utterance, the object is a group rather than a solver, and the ranking is scored against an expert answer, so error is a number rather than a construct anyone has to code. That makes it the cheapest project in the cluster to measure and the one furthest from the cluster’s own definition problem. It belongs here because it is the only one that already has both a human corpus and a synthetic one over the same task, which is what makes the comparison possible at all.

Three terms on the page and in the paper do not carry the same value, and all three are recorded below rather than reconciled. One of them changes what the survival thresholds mean.

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1 Active

1.1 The task and the score

winter survival task

“A plane goes down in a frozen forest and twelve salvaged items have to be ranked by how much they matter to staying alive”

website, Small Groups, opening

Definition: A group decision-making exercise. Each person ranks the salvaged items alone, then the group deliberates to a single ranking, under a time limit. Both rankings are scored against a human expert ranking.

Distinguishes it from: A hidden-profile task, where the group has to pool information no single member holds. Here every member starts with the same information and the same items. What varies between members is judgement, so the whole of the group effect has to come from talk and from averaging, not from disclosure.

Operational test: Is there an expert ranking? Without one there is no error to measure, and the task becomes a study of agreement rather than of accuracy.

In this project: The task paradigm is Humphreys, Johnson and Johnson 1982. The human data is the Group Affect and Performance corpus of Braley and Murray 2018. The agent groups run the same task under the free-form conversation algorithm.

Twelve items or fifteen, and why the number matters

The page names twelve: a lighter, newspaper, a compass, gauze, a knife, an air map, canvas, a shirt, whisky, chocolate, rope, an ax.

The paper describes the paradigm with fifteen: “Participants are presented with these 15 items and must work in small groups to discuss and rank each item according to its importance for their survival in that situation.”

The exercise circulates in both a twelve-item and a fifteen-item form, so neither number is a mistake. The consequence is arithmetic and is not optional. The score subtracts a sum of absolute rank differences from 100, and the largest that sum can be is $\lfloor n^2/2 \rfloor$: 72 at twelve items, 112 at fifteen. So the floor of the scale is 28 in one version and -12 in the other, and the survival bands, 50 and above survives, 40 to 49 risks frostbite, 30 or below is serious danger, sit at different points on the two scales. Any number quoted from one version cannot be read against the other without saying which. Fix the item count before quoting a score anywhere.

expert ranking

“There is an expert ranking, so error is measurable.”

website, Small Groups, opening

Definition: The reference ordering of the items, supplied with the exercise, against which every individual and group ranking is scored.

Distinguishes it from: A consensus ranking, which is what the group produces. The two are compared, never conflated, and the group can converge firmly on a wrong one.

Operational test: Would the score change if the group agreed more? No. The score is distance to the expert, and agreement enters only through which ranking gets submitted.

In this project: It is the anchor for both scores below and it is the reason this task was chosen over an open deliberation task.

absolute individual score and absolute group score

Definition: Two numbers on one scale. The individual score is

$$\text{AIS} = 100 - \sum_i |\text{Rank}_{\text{individual}}(i) - \text{Rank}_{\text{expert}}(i)|,$$

and the group score is the same expression with the group’s ranking in place of the individual’s. Groups at 50 and above survive; between 40 and 49 a member might get frostbite; from 30 to 39 at most three members survive; at 30 and below the group is in serious danger.

Distinguishes it from: A rank correlation. Sum of absolute differences penalises a single item moved far more heavily than a correlation does, which is the intended behaviour when the items are survival goods and one of them is the lighter.

Operational test: Does the metric change if two adjacent items swap at the bottom of the list? Under this score it changes by two, the same as a swap at the top. Positional weighting is not in the measure.

In this project: Random performance was fitted as a Gaussian with mean 15.34 and standard deviation 12.71, $R^2 = 0.95$, and that is the baseline any group score is read against.

1.2 What the discussion does

pooling

“Pooling already helps before anyone says a word”

website, Small Groups, what the discussion does

Definition: The ranking obtained by averaging the members’ independent rankings, with no conversation. The statistical benefit of several noisy estimates of one quantity.

Distinguishes it from: The group ranking after discussion. These are the two numbers that have to be kept apart, because the interesting quantity is the difference between them and not either one on its own.

Operational test: Could the improvement have been obtained by a clerk with the four sheets of paper and no room? If yes, the talk did nothing that averaging would not have done.

In this project: The page carries three error figures side by side for exactly this reason: how wrong the average member was alone, how wrong the simple pool is, and how wrong the group is after arguing. The paper’s version of the same comparison is the average individual score before discussion set against the group score after it.

free-form conversation

“we propose a mechanism that enables free-form discussions among an arbitrary number of agents without imposing predefined interaction rules”

Definition: A conversation in which no speaking order is imposed and the next speaker is self-selected. Each listening agent monitors the history and decides for itself whether to claim the floor or stay silent.

Distinguishes it from: Round-robin prompting, which is what most multi-agent setups do and which fixes turn order in advance. Turn allocation cannot be predetermined in human conversation, so a fixed order is a design choice that throws away the thing being measured.

Operational test: Can the same agent speak twice in a row? Under free-form it can, because the speaker keeps the floor until someone claims it. Under round-robin it cannot.

In this project: The algorithm is the first of the paper’s three contributions. If more than one agent claims the floor, one is chosen at random. If no agent recognises an obligation to speak and no consensus has been reached, the conversation ceases and the group task ends in failure.

conversation floor

Definition: Possession of the right to speak. Held by whoever is speaking, released voluntarily or taken by a claim from a listener.

Distinguishes it from: The turn, which is what a speaker produces while holding the floor. One holding of the floor can span several utterances.

Operational test: Ask who would speak next if nobody intervened. Under a floor model the answer is the current speaker. Under a turn model there is no answer without an order.

In this project: The floor action is one of two reasoning sub-actions each agent performs after every utterance it observes. The other is the ranking update.

ranking update

Definition: The step in which an agent reconsiders the propositions made in the conversation so far, integrates them, and rewrites its own ranking of the items, before deciding whether to speak.

Distinguishes it from: The utterance. The ranking update runs after every observed utterance, including the ones the agent does not respond to, so an agent can be persuaded without ever saying so.

Operational test: Does the agent’s private state change when it stays silent? If not, silence carries no information and the conversation is only its transcript.

In this project: It is what makes a group ranking recoverable at any point in the conversation rather than only at the end, and it is the reason the animation on the page can show the talk and the ranking side by side.

1.3 What the corpus is measured on

group action annotation

Definition: Four labels applied to each utterance: proposal, agreement, disagreement, confirmation. One utterance may carry more than one.

Distinguishes it from: Sentiment, which is a separate annotation on the same utterance.

An utterance can be a disagreement and positive at once, and the paper reports agent groups combining agreement and disagreement in single statements more often than people do.

Operational test: Would the label change if the content were about something else? If yes it is not a group action, it is a topic code. These four are about the decision process, not the decision.

In this project: Applied to the human corpus by hand and to the synthetic corpus automatically, with a network fine-tuned on 60% of the annotated human corpus reaching 72.4% agreement with the human annotation on the remaining 40%.

sentiment annotation

Definition: Two counts per conversation, positivity and negativity, being the number of utterances annotated positive and negative.

Distinguishes it from: Satisfaction, which is self-reported after the task on five-point scales and is a judgement about the meeting rather than about an utterance. The two can and do move in opposite directions.

Operational test: Is the label on the utterance or on the person? Sentiment is on the utterance. Satisfaction is on the person.

In this project: Automated with a distilled transformer fine-tuned on 60% of the human corpus, reaching 81.3% agreement on the remaining 40%.

airtime proportion

Definition: The words uttered by one speaker divided by the total word count of the conversation.

Distinguishes it from: Turn count, which counts occasions rather than volume. A member who interjects often and briefly has many turns and little airtime.

Operational test: Does an even distribution mean an even conversation? Not by itself, which is why the distribution is reported rather than its mean.

In this project: The distributions separate most at group size four. More than 40% of agents occupied between 20% and 30% of the airtime, against 33% of humans, and the paper’s example timeline shows one human group dominated by a single member.

meeting length

Definition: Conversation length measured in words rather than in minutes.

Distinguishes it from: Elapsed time, which is not comparable across the two populations: the agents are not embedded in the world and produce text as fast as the hardware allows. Back channels, coughs, nods and unclear fragments, are removed from the human corpus so the two counts mean the same thing.

Operational test: Would the number change if the same conversation were held slowly? Under a word count it would not. That is the reason for the choice.

In this project: Agent meetings are shorter than human meetings by a wide margin, $F(1, 147) = 355.7$, $p < 0.0001$. Human meeting length grows with group size; agent meeting length does not.

human-likeness

Definition: The property being measured of the synthetic corpus: whether agent group behaviour

resembles human group behaviour on performance and efficiency, affect and satisfaction, and group action and airtime.

Distinguishes it from: Response quality, which is what conversational dialogue systems are usually evaluated on, and which is a property of one utterance at a time. This project’s second contribution is moving the question from the utterance to the group.

Operational test: Could the metric be computed on a single agent talking to itself? If yes it is not a group-level metric and does not belong in this set.

In this project: It is the reason the same annotation scheme is applied to both corpora rather than each being described in its own terms.

2 Working

human corpus

Definition: The Group Affect and Performance corpus, Braley and Murray 2018. Twenty-eight groups, eighty-four participants: six groups of two, sixteen of three, six of four. Speaker-level demographics and questionnaire answers, utterance-level transcription, timestamps, sentiment and decision annotation.

Distinguishes it from: The synthetic corpus generated by the agent groups over the same task, at the same three group sizes, with the same four decision labels applied.

back channel

Definition: A short listener signal that does not claim the floor: a cough, a nod, an unclear fragment.

Distinguishes it from: An interjection, which does claim it. Back channels are excluded from the human word counts because the text-based agents produce no equivalent, and their presence would make the length comparison unfair.

peer evaluation

Definition: Five post-task statements answered on five-point scales: time expectation, worked well together, time management, efficiency, quality of work. Satisfaction is the group average over the members’ averages.

Distinguishes it from: The conversation-level sentiment counts. Agents score lower than people on time management and on efficiency, and higher on time expectation, worked well together and quality of work, all at $p < 0.001$ except worked well together at $p < 0.05$.

turn-taking

Definition: The organisation of transitions between speakers.

Distinguishes it from: Speaking order. Transitions can happen with or without gaps, order can vary, and the relative distribution of turn allocation cannot be fixed in advance, which is the finding the free-form algorithm is designed around.

3 Peripheral

dialogue act classification

The task the decision annotation is automated as. The reference corpus is the Switchboard dialogue act corpus and the annotation manual behind it.

episodic memory

Named as absent by design. The agents carry the conversation history as a list of speaker and text pairs across prompts, which the paper likens to working memory, and nothing across sessions, because each group performs the task once.

model version

The 0613 build of a 32k-context frontier model, fixed for reproducibility. Utterance length is capped at forty words, the longest utterance in the human corpus. An auxiliary attribute records whether a speaker is addressing one partner or the whole group.

4 The result, stated the way the evidence supports

The page states three differences and a conclusion. The paper’s numbers support them at different strengths, and the differences are worth carrying, because the page’s sentence is the one that would go in a talk.

“Agent groups produced more disagreements, more complex statements, and a marked preference for positive statements compared with people. So the advantage doesn’t come from being agreeable and converging quickly. It comes with more argument, not less.”

website, Small Groups, groups of language models

Claim on the page	What the paper reports
Outperformed the human groups	One-way ANOVA on human against agent, across three group sizes: $F(1, 147) = 5.121, p < 0.05$
Gained more from discussion	Improvement is higher than humans at group size two and group size four, both at $p < 0.1$. Group size three is not among them
More disagreements	Agents express disagreement significantly more often than humans, and make proposals more often, especially combined with agreement, disagreement or confirmation
More complex statements	Reported qualitatively, as statements that combine agreement and disagreement in one utterance. There is no separate complexity metric in the measure set
Preference for positive statements	Positivity is higher but not separated: $M = 15.218, SD = 7.322$ for agents against $M = 13.785, SD = 11.767$ for humans. Negativity is separated and lower: $M = 3.075, SD = 2.046$ against $M = 5.285, SD = 6.759, F(1, 147) = 9.29, p < 0.01$

The preference for positive statements is a shortage of negatives

The page reads the affect result as a preference for positive statements. The table it rests on separates only the negative count. Both populations produce more positive utterances than negative ones, and the agents are not distinguished by producing more positives; they are distinguished by producing fewer negatives.

The distinction matters for the mixed-team question the page ends on. An agent that says more nice things and an agent that withholds criticism behave the same way in a friendly room and differently in a hostile one. If the second reading is right, then the willingness to disagree that the page counts as an advantage is already sitting next to a reluctance to say anything negative, in the same corpus, from the same agents.

The safe sentence for a talk is the one the abstract uses: more disagreements, complex statements, and a propensity for positive statements. The strong version, that agents prefer positive statements, needs the positivity difference to hold up and it does not, yet.

5 What this project leaves open

The page ends on the question, and it is the right one to end on.

“whether an agent’s willingness to disagree survives contact with a human who outranks it”

website, Small Groups, groups of language models

The paper’s own limitation points the same way from the other side: the human groups consist of different people with different background knowledge, biases and preferences, while the agent groups can be less diverse. Both sentences are about the same missing condition, a mixed group, and neither study has run one. The efficiency half of the index question is untouched: nothing in either corpus identifies which exchanges did the work, only that the ranking improved after them.

Two further openings, both cheap. The free-form algorithm terminates a conversation in failure when no agent recognises an obligation to speak, which is an exit condition chosen rather than derived, and the cluster’s control-layer problem is exactly about exit conditions. And the annotation of the synthetic corpus was automated against a network fitted to the human corpus, which means the agent conversations are described in the vocabulary of human conversations by construction. Anything the agents do that the human labels have no word for is invisible to this measure set.

6 Sources

- Every quoted sentence marked *website* is from the Small Groups page under `3-year-agenda/mental-world-modeling/ problem-solving/`, or from the Problem-Solving index above it.
- Every number, every statistic, and every quoted sentence marked *paper* is from Du, Rajivan and Gonzalez, “Large Language Models for Collective Problem-Solving: Insights into Group Consensus Decision-Making”, Proceedings of the 46th Annual Conference of the Cognitive Science Society, 46 (2024), 3011 to 3018.
- The task paradigm is Humphreys, Johnson and Johnson 1982. The human corpus is Braley and Murray 2018.

- Shared cluster terms, grain, elicitation, the control layer, reliability, typing another agent, are defined once in `mwm_shared.tex` and are not repeated here.